

Strategic Portfolio Design Analyst Role- Risk Case Study

Take Home Case Study

Quantitative Asset Allocation

Context

You are advising an institutional multi-asset portfolio team. The team is reviewing whether its current long-term asset allocation could be improved using a more systematic and evidence-based portfolio construction process.

You have been provided with a dataset of monthly asset class returns. Your task is to design, justify, and evaluate an asset allocation recommendation for a long-term institutional investor.

There is no single correct answer. We are interested in your investment reasoning, analytical judgment, coding approach, and ability to communicate trade-offs clearly.

You may use any tools you normally would, including AI assistance, but you should be prepared to explain every design choice, assumption, and analytical step in a follow-up discussion.

Data Provided

You are given monthly returns for a set of broad asset classes, along with a cash or risk-free return series. Assume the asset classes are investable proxies for broad public-market exposures.

Assignment

Your task is to recommend a strategic asset allocation for a long-term institutional investor.

The investor is seeking a diversified portfolio that can earn a return above cash over the long term while managing downside risk and remaining suitable for institutional governance.

You should decide how to use the data, what assumptions to make, what portfolio construction approach to apply, whether any constraints are appropriate, and how to evaluate the resulting allocation.

Your recommendation should be supported by analysis, but it should also reflect investment judgment. You should not assume that the historical sample alone provides a complete answer.

Required Analysis

Please review the data, construct a recommended allocation, and evaluate the result.

Your work should address the following areas.

1. Data Assessment

Review the return data before constructing the portfolio.

Discuss any data issues, assumptions, or limitations that are relevant to your analysis.

2. Portfolio Construction Approach

Develop a portfolio construction approach that you believe is suitable for the investor.

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Explain the objective of your approach, the key assumptions you made, and why the resulting allocation is appropriate.

You may apply constraints if you believe they are appropriate. If you do, please explain what constraints you used and why.

3. Evaluation

Evaluate your recommended allocation using methods you believe are appropriate for judging a long-term institutional portfolio.

Explain how the portfolio performs, how it compares to relevant alternatives, and whether you believe the result is robust enough to inform a real allocation decision.

Recommendation

Provide a clear final recommendation.

Your recommendation should include the final allocation, the rationale for the allocation, key limitations, and any additional work you would do with more time or better data.

The written submission does not need to be lengthy. We are more interested in the quality of your analysis and your ability to discuss the work in person than in a polished report.

Deliverables

Please submit:

1. A brief written summary, no more than 2–3 pages.
2. Your code in Python.
3. A short README explaining how to run the code.
4. Your recommended asset allocation weights.
5. Any supporting exhibits you believe are useful.

The written summary should be concise and focused on the main decisions, results, and trade-offs. Most of the discussion will take place during the follow-up interview.

Follow-Up Discussion

In the interview, be prepared to discuss your approach in detail, including the choices you made, alternatives you considered, limitations of your analysis, and how you validated your results.

You should also be prepared to walk through your code and explain what role, if any, AI played in your process.